

Problem1.3

April 19, 2026

0.1 Problem 1.3 — Extremal curves

0.1.1 1.3

Using the Euler-Lagrange equation in the Fundamental Theorem of Calculus of Variations as a necessary condition for an optimal curve:

$$\frac{\partial}{\partial x} \left(\frac{1}{2} \dot{x}(t)^2 + 5x(t)\dot{x}(t) + x(t)^2 + 5x(t) \right) - \frac{d}{dt} \frac{\partial}{\partial \dot{x}} \left(\frac{1}{2} \dot{x}(t)^2 + 5x(t)\dot{x}(t) + x(t)^2 + 5x(t) \right) = 0$$

$$5\dot{x}(t) + 2x(t) + 5 - \frac{d}{dt} [\dot{x}(t) + 5x(t)] = 0$$

$$5\dot{x}(t) + 2x(t) + 5 - \ddot{x}(t) - 5\dot{x}(t) = 0$$

$$\boxed{\ddot{x}(t) - 2x(t) = 5}$$

General solution for $A, B \in \mathbb{R}$:

$$x^*(t) = A \cosh(\sqrt{2}t) + B \sinh(\sqrt{2}t) - \frac{5}{2}$$

with BC $x^*(0) = 1$ and $x^*(1) = 3$.

$$x^*(0) = A - \frac{5}{2} = 1 \quad \Rightarrow \quad A = \frac{7}{2}$$

$$x^*(1) = \frac{7}{2} \cosh(\sqrt{2}) + B \sinh(\sqrt{2}) - \frac{5}{2} = 3 \quad \Rightarrow \quad B = \frac{\frac{11}{2} - \frac{7}{2} \cosh(\sqrt{2})}{\sinh(\sqrt{2})}$$

$$\boxed{x^*(t) = \frac{7}{2} \cosh(\sqrt{2}t) + \frac{11 - 7 \cosh(\sqrt{2})}{2 \sinh(\sqrt{2})} \sinh(\sqrt{2}t) - \frac{5}{2}}$$

Please see plot.

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[1]: import numpy as np
import matplotlib.pyplot as plt

A = 7 / 2
B = (11/2 - (7/2) * np.cosh(np.sqrt(2))) / np.sinh(np.sqrt(2))

def x_star(t):
    return A * np.cosh(np.sqrt(2) * t) + B * np.sinh(np.sqrt(2) * t) - 5/2

t = np.linspace(0, 1, 500)

fig, ax = plt.subplots(figsize=(7, 4))
ax.plot(t, x_star(t), lw=2)
ax.scatter([0, 1], [1, 3], color='k', zorder=5)
ax.set_xlabel(r'$t$')
ax.set_ylabel(r'$x^*(t)$')
ax.set_title(r'Extremal curve for $J(x)$')
ax.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```

